

Homework — 1.1.2 Types of Processor

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.1.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. Explain why the **Von Neumann** architecture can suffer from a performance **bottleneck**, and state how the **Harvard** architecture avoids it. **[3]** (AO1)

2. State whether each scenario is better suited to **Von Neumann** or **Harvard** architecture, with a brief reason. **[2]** (AO2)

- A general-purpose laptop running many different applications: _____
- A digital signal processor (DSP) inside a low-cost audio device: _____

3. A smartphone manufacturer uses a **RISC (ARM)** processor rather than a **CISC (x86)** processor. Give **two** reasons, linked to a smartphone, why RISC is the better choice. **[2]** (AO2)

4. In a **CISC** processor the complexity sits in the **hardware**; in a **RISC** processor it sits in the **software/compiler**. Explain **one** consequence of this difference for **pipelining**. **[2]** (AO1)

5. Describe what is meant by **data parallelism**, and explain why a **GPU** is well suited to it. **[3]** (AO1)

6. A video-editing application is rewritten to spread its work across **all the cores** of a multicore CPU, yet the speed-up is far less than the number of cores would suggest. Explain **two** reasons why the speed-up is limited. **[2]** (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

7. Many modern CPUs are described as a **modified (hybrid) Harvard** architecture. Explain what this means. **[2]**

8. State **two** factors (other than number of cores) that affect the performance of a CPU, and for each give its effect. **[2]** (*retrieval from 1.1.1*)

9. Name the register that holds the **current instruction** while it is being decoded and executed. **[1]** (*retrieval from 1.1.1*)

10. Give **one** non-graphics (GPGPU) application of a GPU. [1]
